SCADA-i (Supervisory Control and Data Acquisition Intelligence)

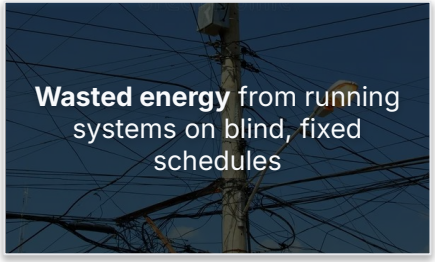
UMHackathon 2026 Submission

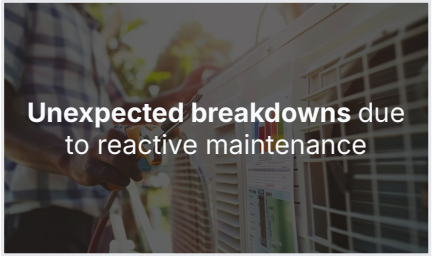
Drowning in Data, Starving for Automation

Modern facilities generate massive amounts of IoT data, yet facility managers still rely on tedious manual analysis to turn raw data into valuable actions (e.g. generating ESG report, scheduling maintenance, managing room bookings).

This friction is caused by traditional monitoring dashboards **also known as SCADA**, which only displays raw static numbers. Forcing human operators to constantly stare at screens to interpret trends, catch inefficiencies, and manage daily operations.

Which leads us to,





> ***"In an era of smart buildings, data is no longer the problem, it is the failure to act on it timely and intelligently."***

The Solution: SCADA-i

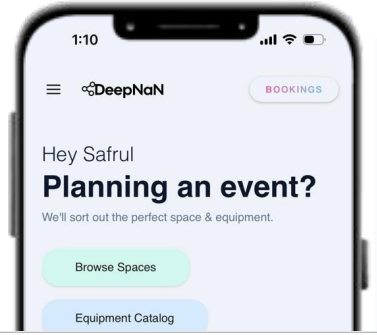
Say goodbye to outdated monitoring dashboards — welcome **SCADA-i**.
An autonomous, supervisory multi-agent system that acts on data in real time.

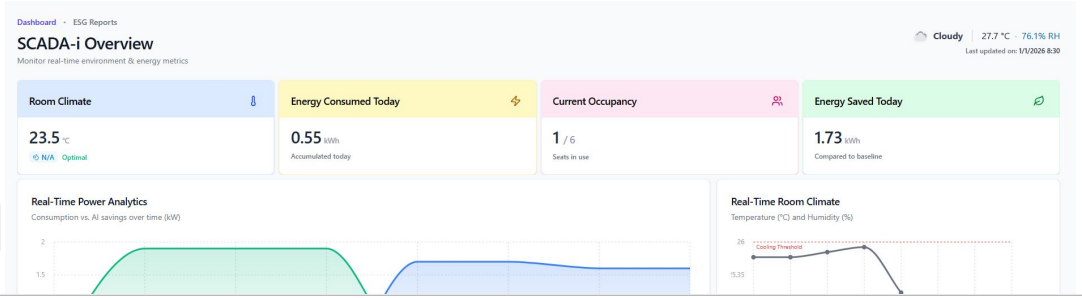
REDUCE ENERGY
WITHOUT SACRIFICING COMFORT

UNIFIED ECOSYSTEM
HVAC, BOOKINGS & ESG

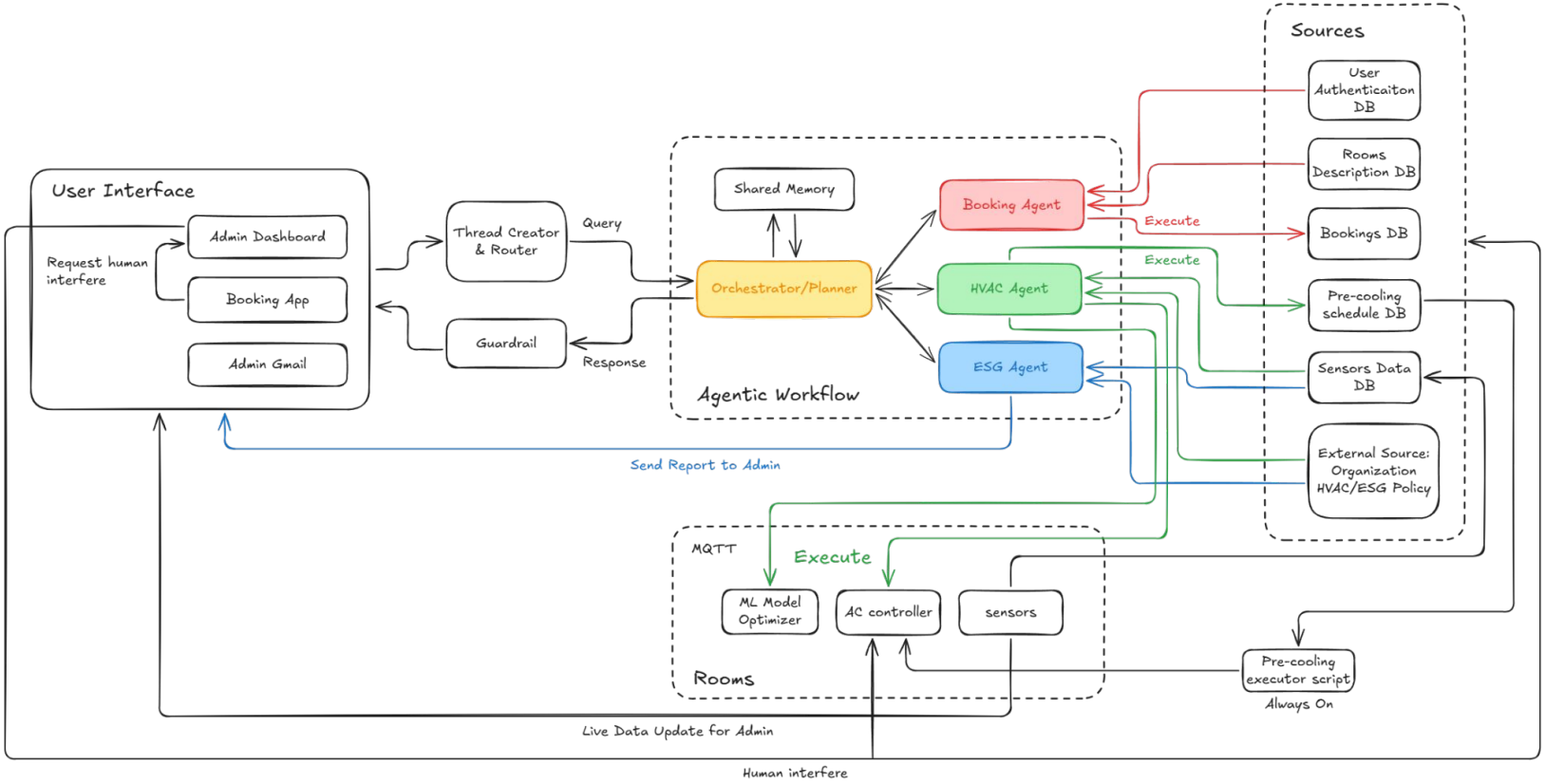
DETECT ANOMALIES
BEFORE THEY BECOME COSTLY

MANAGE BUILDINGS
USING Z.AI ILMU-GLM-5.1





System Architecture & Operational Workflow



How the Agents Work Together

Orchestrator (Master Agent)

Routes user intent to specialized agents and coordinates multi-agent workflows
Satisfies user needs while optimizing ESG at the same time

Booking Agent

Maintains an up-to-the-second understanding of room capacities, features, and availability.

Autonomously handles all user reservations, modifications, and cancellations.

HVAC Agent

Continuously monitors real-time room conditions, occupancy and user requests.

Dynamically controls temperature and airflow optimized for comfort, energy usage and system efficiency.

ESG Agent

Audits building operations by analyzing HVAC compliance and space utilization.

Automatically generates ESG reports that summarize a building's environmental impact.

Scalable Benefits

Reclaiming **40% of wasted energy** across standard 35m² classrooms and corporate meeting rooms.

Saving
RM 6,700/yr

14.7 MWh of energy reclaimed per 4-room cluster.

Covering initial capital investment in less than 3 months.

Avoiding
11.4 tons of CO_2

Directly aligns with UN SDGs and corporate ESG mandates.

Equivalent to planting ~550 trees every year per cluster.

Saving
40+ hrs/month

Eliminates daily assessment, analysis and reporting.

Z.AI GLM orchestrates specialised AI Sub-Agents end-to-end.

No wasted energy. No admin frustration. Better alignment. — all within **one closed-loop system.**

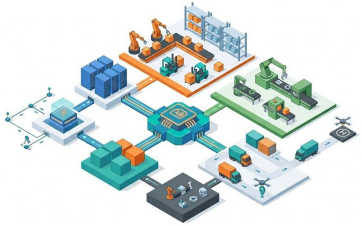
Implementation Roadmap (Tentative)

Stage 1: System Understanding & Integration

- Analyze existing building systems
- Identify integration points for AI-driven automation.

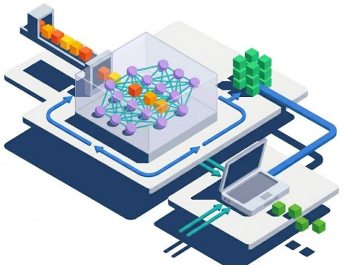
Stage 3: Model Training & System Testing

- Develop and test multi-agent AI and orchestration logic.
- Validate system decisions and workflows.



Stage 2: Data & Infrastructure Setup

- Deploy sensors and IoT infrastructure to collect real-time data.



Stage 4: Deployment & Autonomous Operations

- Deploy with monitoring, automation and optimization.
- Supervise autonomous building workflows.

That’s all,
Thank you for listening!

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